Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 1 of 7 · Question 4 part b · 3 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

4(a)	$\overrightarrow{DC} = (11\mathbf{i} - p\mathbf{j}) - (4\mathbf{i} - 2p\mathbf{j}) = 7\mathbf{i} + p\mathbf{j} = \overrightarrow{AB}$ OR: $\overrightarrow{BC} = (11\mathbf{i} - p\mathbf{j}) - (7\mathbf{i} + p\mathbf{j}) = 4\mathbf{i} - 2p\mathbf{j} = \overrightarrow{AD}$	M1A1	
	Parallel and equal in length ∴ Parallelogram	A1cso	(3)
(b)	$\overrightarrow{BD} = (4\mathbf{i} - 2p\mathbf{j}) - (7\mathbf{i} + p\mathbf{j}) = -3\mathbf{i} - 3p\mathbf{j}$ (or $3(-\mathbf{i} - p\mathbf{j})$ oe	B1	
	$\sqrt{9 + (3p)^2} = 3\sqrt{10}$ $(\Rightarrow 9 + 9p^2 = 90)$	M1	
	$p = \pm 3$	A1	(3)
(c)	$(\pm)\frac{1}{3\sqrt{10}}(-3\mathbf{i} - 9\mathbf{j})$ oe	B1ft	(1)
			[7]

Accept column vectors throughout.

(a)

M1 Attempt $\pm\underset{\sim}{DC}$ or $\pm\underset{\sim}{BC}$ using the difference of 2 appropriate vectors in component form.

A1 Show that $\pm\underset{\sim}{DC}$ or $\pm\underset{\sim}{BC} = \pm\underset{\sim}{AB}$ or $\pm\underset{\sim}{AD}$

A1cso Suitable conclusion with reason from correct working.
One pair of vectors only needed if reason is "parallel and equal". Both pairs needed if reason is "2 pairs of sides parallel/equal".

(b)

B1 For a correct $\underset{\sim}{BD}$ or $\underset{\sim}{DB}$. No simplification needed.

M1 Use the given length of $\underset{\sim}{BD}$ with the length of their $\underset{\sim}{BD}$ to form an equation

A1 Obtain correct values for p. Both needed.

(c)

B1ft Use their positive value for p to obtain a unit vector (no simplification needed)

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2020 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 2 of 7 · Question 1 part b · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 2 of 7 · Question 1 part b · 1 mark

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Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 2 of 7 · Question 1 part b · 1 mark

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

International GCSE Further Pure Mathematics – Paper 2R mark scheme

Question number	Scheme	Marks
1(a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$ or $6\mathbf{i} + 8\mathbf{j} = \overrightarrow{OB} - (3\mathbf{i} - 2\mathbf{j})$ oe $9\mathbf{i} + 6\mathbf{j}$	M1 A1 [2]
(b)	$\sqrt{6^2 + 8^2}$ *or* 10 (from Pythagorean triple)	B1 [1]
(c)	$(\pm)\frac{1}{"10"}(6\mathbf{i} + 8\mathbf{j})$	M1 A1 [2]
		Total 5 marks

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 3 of 7 · Question 4 part a · 4 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

4(a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = 5\mathbf{i} + 7\mathbf{j} + a\mathbf{i} + 16\mathbf{j} = (5+a)\mathbf{i} + 23\mathbf{j}$ $\left(5\sqrt{29}\right)^2 = (5+a)^2 + 23^2 \Rightarrow 5+a = \pm\sqrt{196} \Rightarrow a = 9, -19$	M1A1 M1A1 [4]
(b)	$\overrightarrow{AB} = "9"\mathbf{i} + 16\mathbf{j} \Rightarrow \left\|\overrightarrow{AB}\right\| = \sqrt{"9"^2 + 16^2} = \sqrt{337}$ Unit vector: $\frac{1}{\sqrt{"337"}}("9"\mathbf{i} + 16\mathbf{j})$ oe.	M1 A1 [2]
		Total 6 marks

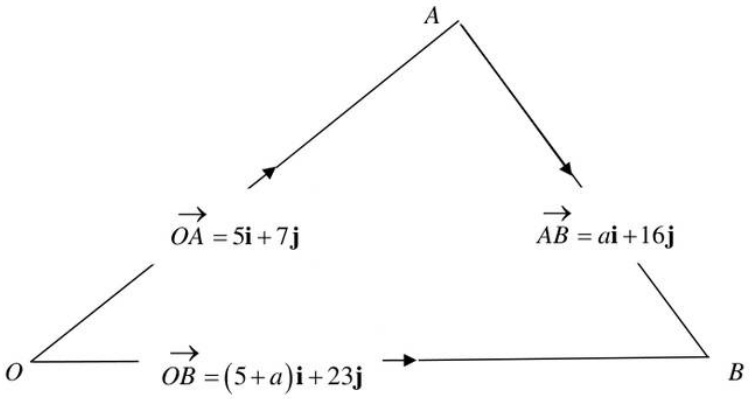
Part	Mark	Notes
(a)	M1	For the correct vector statement for $\overrightarrow{OB}$ For example, accept $\overrightarrow{AB} = \overrightarrow{A0} + \overrightarrow{OB}$ This mark can be implied by a correct vector for $\overrightarrow{OB}$
	A1	For the correct vector in terms of a [simplified or unsimplified].
	M1	For using Pythagoras theorem with $\left(5\sqrt{29}\right)$ and their vector for $\overrightarrow{OB}$ and solving the equation to find **two** values of a
	A1	For the **two** correct values. $a = 9, -19$ seen in their working
(b)	M1	For finding $\lvert AB\rvert$ by using a correct Pythagoras and writing down the unit vector where $\overrightarrow{AB} = k\mathbf{i} + 16\mathbf{j}$ where k is a **positive** value. Award for $\frac{"9"\mathbf{i} + 16\mathbf{j}}{\sqrt{"9"^2 + 16^2}}$ $\sqrt{9^2 + 16^2}$ can be implied by sight of $\sqrt{337}$ If they have an incorrect value for a, full substitution using Pythagoras theorem must be seen for the award of this mark.
	A1	For the correct unit vector. Accept any correct equivalent unit vectors. For example; $\frac{\sqrt{"337"}}{"337"}("9"\mathbf{i} + 16\mathbf{j})$ or $\frac{"9"\mathbf{i}}{\sqrt{"337"}} + \frac{16\mathbf{j}}{\sqrt{"337"}}$ Accept decimal answers. Eg., $0.054("9"\mathbf{i} + 16\mathbf{j})$ for awrt 0.054 Or $0.49\mathbf{i} + 0.87\mathbf{j}$ or better. Allow $\pm\frac{1}{\sqrt{"337"}}("9"\mathbf{i} + 16\mathbf{j})$

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 3 of 7 · Question 4 part a · 4 marks

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Useful Sketch



Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 3 of 7 · Question 4 part a · 4 marks

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Question	Scheme	Marks

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 4 of 7 · Question 3 part a · 4 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

3 a	$\left\lvert\overrightarrow{OA}\right\rvert=\sqrt{p^2+16}$ and $\left\lvert\overrightarrow{OB}\right\rvert=\sqrt{4p^2+4p+2}$	
	$\sqrt{2}\left\lvert\overrightarrow{OA}\right\rvert=\left\lvert\overrightarrow{OB}\right\rvert\Rightarrow 2p^2+32=4p^2+4p+2$	M1
	$2p^2+4p-30=0\Rightarrow p^2+2p-15=0$	M1
	$(p+5)(p-3)=0$	M1
	$p=3$	A1 (4)
b	$\overrightarrow{AB}=-'3'\mathbf{i}+4\mathbf{j}+\mathbf{i}+(2\times'3'+1)\mathbf{j}=-2\mathbf{i}+11\mathbf{j}$	M1 A1
	$\left\lvert\overrightarrow{AB}\right\rvert=\sqrt{'4'+'121'}\ \left[=5\sqrt{5}\right]$	M1
	$'\left[\frac{1}{5\sqrt{5}}\right]"(-2\mathbf{i}+11\mathbf{j})'$	dM1
	$(\pm)\ \frac{1}{5\sqrt{5}}(-2\mathbf{i}+11\mathbf{j})$	A1 (5)
		Total 9 marks

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 4 of 7 · Question 3 part a · 4 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	**M1**	For use of $\sqrt{2}\left\|\overrightarrow{OA}\right\| = \left\|\overrightarrow{OB}\right\|$ i.e., $\sqrt{2}\times\sqrt{p^2+(-4)^2} = \sqrt{1+(2p+1)^2} \Rightarrow \left(\sqrt{2}\times\sqrt{p^2+16} = \sqrt{4p^2+4p+2}\right)$ They may find $\left\|\overrightarrow{OA}\right\|$ and $\left\|\overrightarrow{OB}\right\|$ separately. Award when combined with $\sqrt{2}$ and condone arithmetical slips.
	M1	For forming a 3TQ in any order. [The correct 3TQ is $2p^2+4p-30=0$ or $p^2+2p-15=0$]
	M1	For a correct attempt to solve **their** 3TQ by any valid method. They must reach a value of p for this mark.
	A1	For $p=3$ If they also give $p=-5$ without evidence of rejecting this solution, withhold the A mark
(b)	**M1**	For the vector statement $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$ o.e. This can be implied by sight of $\overrightarrow{AB} = '-2'\mathbf{i} + '11'\mathbf{j}$ or $\overrightarrow{AB} = \begin{pmatrix} '-2' \\ '11' \end{pmatrix}$ If there is no vector statement you must check their vector for substitution of their p.
	A1	For the correct $\overrightarrow{AB}$ (allow unsimplified) $\left(\overrightarrow{AB} = -2\mathbf{i}+11\mathbf{j}\right)$ and also allow $\overrightarrow{AB} = \begin{pmatrix} -2 \\ 11 \end{pmatrix}$ Award for sight of $-2\mathbf{i}+11\mathbf{j}$ only.
	M1	For using Pythagoras theorem on their $\overrightarrow{AB}$ i.e., $\sqrt{('-2')^2+'11'^2}$
	dM1	For correct method to find a unit vector using their values and their $\overrightarrow{AB}$ NB: this mark is dependent on the previous M mark
	A1	For $\frac{1}{5\sqrt{5}}(-2\mathbf{i}+11\mathbf{j})$ oe $\left[\text{Allow } -\frac{1}{5\sqrt{5}}(-2\mathbf{i}+11\mathbf{j})\right]$ **OR** $\overrightarrow{AB} = \pm\frac{1}{5\sqrt{5}}\begin{pmatrix} -2 \\ 11 \end{pmatrix}$

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 4 of 7 · Question 3 part a · 4 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks
	$O_1A = O_2A = O_1O_2 = 6$ so triangle AO_1O_2 is equilateral	B1

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 5 of 7 · Question 3 part a · 4 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

3(a)	$\overrightarrow{OA} = \overrightarrow{OB} - \overrightarrow{AB} = 7i + 2aj - (i + 3aj) = 6i - aj$ $\left\|\overrightarrow{OA}\right\| = \sqrt{6^2 + a^2} = 3\sqrt{5} \Rightarrow 6^2 + a^2 = 45 \Rightarrow a^2 = 9 \Rightarrow a = 3$	M1A1 M1A1 [4]
(b)	$\overrightarrow{OA} = 6i - '3'\, j$ {Unit vector parallel to $\vec{OA}$ is} $\{\pm\}\frac{1}{3\sqrt{5}}(6i - 3j)$ oe ,isw	B1ft B1 [2]
		Total 6 marks

Part	Mark	Notes
(a)	M1	For the correct vector statement with vector $\vec{OA}$ or $\vec{AO}$ e.g. $\overrightarrow{OA} = \overrightarrow{OB} - \overrightarrow{AB}$ or $\left\{\overrightarrow{OA} =\right\} 7i + 2aj - (i + 3aj)$ oe
	A1	For the correct simplified or unsimplified vector $\vec{OA}$ **or** $\vec{AO}$
	M1	For using the correct Pythagoras theorem on their $\vec{OA}$ or $\vec{AO}$ with $3\sqrt{5}$, leading to a value for a^2 or a
	A1	For the correct value of a=3 only (must reject a=−3 if present) **NB:A value of a=3 must come from a correct vector of OA with no incorrect work seen.**
(b)	B1ft	For the correct vector for $\vec{OA}$ or $\vec{AO}$ using their a (which a>0), (Substitutes their a value to $\vec{OA} = 6i - aj$ or $\vec{AO} = -6i + aj$
	B1	Correct unit vector $\frac{1}{3\sqrt{5}}(6i - 3j)$ *oe*, can be in either direction, so accept + or − isw

Extra notes:

1. Allow working in column vectors throughout.
2. Condone missing arrow on their vectors.
3. **Special Case for part a:** Correct answer with minimum work, no incorrect working shown
e.g. $\sqrt{6^2 + a^2} = 3\sqrt{5}$ M1A1M1 $\rightarrow 6^2 + a^2 = 45 \rightarrow a^2 = 9 \rightarrow a = 3$ A1

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 5 of 7 · Question 3 part a · 4 marks

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Question	Scheme	Marks

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 6 of 7 · Question 6 part a · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Number		
6 (a)	$\left(\overrightarrow{AB}\right) = \overrightarrow{AO} + \overrightarrow{OB} = -9\mathbf{a} - 12\mathbf{b} + 2\mathbf{a} + p\mathbf{b}$	M1
	$(-7)^2 + (p-12)^2 = 25^2$	M1
	$(p-12)^2 = 576$	
	$p = 12 \pm \sqrt{576}$	M1
	$p = -12, 36^*$	A1*
		[4]
(b)	$\sqrt{12^2 + 2^2} \Rightarrow \left(\sqrt{148} = 2\sqrt{37}\right)$	M1
	$\pm\frac{1}{\sqrt{37}}(\mathbf{a} - 6\mathbf{b})$ **or** $\pm\frac{\sqrt{37}}{37}(\mathbf{a} - 6\mathbf{b})$	A1
		[2]
(c)	$(\overrightarrow{OC}) = \frac{2}{5}(2\mathbf{a} + 36\mathbf{b})$	M1
	$(\overrightarrow{CA}) = -\frac{2}{5}(2\mathbf{a} + 36\mathbf{b}) + 9\mathbf{a} + 12\mathbf{b}$	M1
	$= \frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$	A1
		[3]
		Total 9 marks

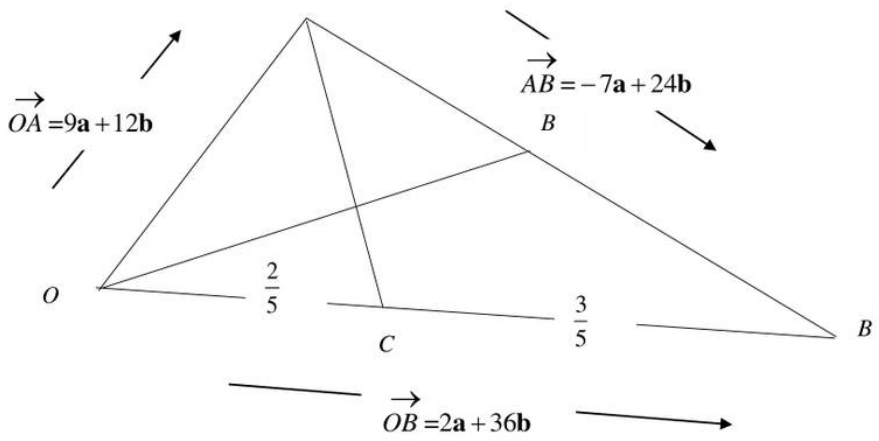
Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 6 of 7 · Question 6 part a · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
6 (a)	M1	For a correct vector path to find $\pm\overrightarrow{AB}$ or $-9\mathbf{a} - 12\mathbf{b} + 2\mathbf{a} + p\mathbf{b}$ or $9\mathbf{a} + 12\mathbf{b} - 2\mathbf{a} - p\mathbf{b}$
	M1	For using Pythagoras' Theorem with th B $\overrightarrow{AB}$
	M1	For any complete and minimally acceptable attempt to solve for p See General Guidance.
	A1*	For both correct values of p – no errors seen. **Note:** This is a show question and the answers are given
(b)	M1	For a corr od to find the magnitude.
O	A1	For a corr $\frac{2}{5}$ vector as shown.
(c)	M1	For a corr or fc $\overrightarrow{C}$
	M1	For the correct vector for $\overrightarrow{CA}$ using their $\overrightarrow{CO} + \overrightarrow{OA}$
	A1	For $\frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$
	ALT: Uses the vector path $\overrightarrow{CA} = \overrightarrow{CB} + \overrightarrow{BA}$	
	M1	For a correct vector for $\overrightarrow{CB} = \frac{3}{5}(2\mathbf{a} + 36\mathbf{b})$ or $\overrightarrow{BA} = 7\mathbf{a} - 24\mathbf{b}$
	M1	For the correct vector for $\overrightarrow{CA}$ using their $\overrightarrow{CB}$ and $\overrightarrow{BA}$ $\overrightarrow{CA} = \frac{3}{5}(2\mathbf{a} + 36\mathbf{b}) + (7\mathbf{a} - 24\mathbf{b})$
	A1	$\overrightarrow{CA} = \frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$

Useful Sketch



Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 6 of 7 · Question 6 part a · 4 marks

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Question number	Scheme	Marks

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 7 of 7 · Question 8 part a.ii · 3 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part a.ii in this mark scheme extract

8 (a) (i)	$\overrightarrow{(AB)} = \overrightarrow{AO} + \overrightarrow{OB}\ (= -4\mathbf{a} - 5\mathbf{b} + 8\mathbf{a} - \mathbf{b}) = 4\mathbf{a} - 6\mathbf{b}$	M1A1 [2]
(ii)	$\left\{\left\|\overrightarrow{AB}\right\| = \sqrt{52}\right\} = 2\sqrt{13}$	B1 [1]
(b)	$\left\|\overrightarrow{OA}\right\| = \sqrt{4^2 + 5^2}\ \left(= \sqrt{41}\right)$ *or* $\left\|\overrightarrow{OB}\right\| = \sqrt{8^2 + 1^2}\ \left(= \sqrt{65}\right)$	M1
	$\left(\sqrt{52}\right)^2 = \left(\sqrt{41}\right)^2 + \left(\sqrt{65}\right)^2 - 2 \times \sqrt{41} \times \sqrt{65} \cos(AOB)$	M1
	$\left\{\text{angle } AOB = \cos^{-1}\left(\dfrac{41 + 65 - 52}{2 \times \sqrt{41} \times \sqrt{65}}\right) = 58.465...\right\}$	
	Area triangle $AOB = \dfrac{1}{2} \times "\sqrt{41}" \times "\sqrt{65}" \times \sin "58.465..."$	M1
	$= 22$ awrt	A1 [4]
ALT 1	$\text{Area} = \dfrac{1}{2}\begin{vmatrix} 0 & 4 & 8 & 0 \\ 0 & 5 & -1 & 0 \end{vmatrix}$ oe	M1
	$= \dfrac{1}{2}\left\|(0 \times 5 + 4 \times (-1) + 8 \times 0) - (0 \times 4 + 5 \times 8 + (-1) \times 0)\right\|$ oe	M1
	$\left\{= \dfrac{1}{2}\left\|-44\right\|\right\} = 22$	M1A1 [4]
ALT 2	Use Heron's formula	
	$\left\|\overrightarrow{OA}\right\| = \sqrt{4^2 + 5^2}\ \left(= \sqrt{41}\right)$ *or* $\left\|\overrightarrow{OB}\right\| = \sqrt{8^2 + 1^2}\ \left(= \sqrt{65}\right)$	M1
	$p = \dfrac{"\sqrt{41}" + "\sqrt{65}" + "\sqrt{52}"}{2}$	M1
	$\text{Area} = \sqrt{p(p - "\sqrt{41}") \times (p - "\sqrt{65}") \times (p - "\sqrt{52}")}$	M1
	$= 22$ exact	A1 [4]
(c)	$\overrightarrow{OC} = \lambda \overrightarrow{OD}\ \{= \lambda(15\mathbf{a} + 10\mathbf{b})\}$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \mu \overrightarrow{AB}\ \{= 4\mathbf{a} + 5\mathbf{b} + \mu(4\mathbf{a} - 6\mathbf{b}) = (4 + 4\mu)\mathbf{a} + (5 - 6\mu)\mathbf{b}\}$	M1
	$4 + 4\mu = 15\lambda$ *and* $5 - 6\mu = 10\lambda \Rightarrow \mu = ...$ *or* $\lambda = ...$	ddM1
	$\mu = \dfrac{7}{26}$ *or* $\lambda = \dfrac{22}{65}$	A1
	$\overrightarrow{OC} = \dfrac{66}{13}\mathbf{a} + \dfrac{44}{13}\mathbf{b}$ or $5\dfrac{1}{13}\mathbf{a} + 3\dfrac{5}{13}\mathbf{b}$ cao	A1 [5]
		Total 12 marks

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Answer 7 of 7 · Question 8 part a.ii · 3 marks

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Part	Mark	Additional Guidance
(a)(i)	M1	For a correct vector path to find $\overrightarrow{AB}$ or $\overrightarrow{BA}$ or for $\pm(-4\mathbf{a}-5\mathbf{b}+8\mathbf{a}-\mathbf{b})$
	A1	For the correct vector $4\mathbf{a}-6\mathbf{b}$
(ii)	B1	For the correct exact simplest magnitude $2\sqrt{13}$
(b)	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For correct substitution into the cosine rule. e.g. $(\sqrt{65})^2=(\sqrt{41})^2+(\sqrt{52})^2-2\times\sqrt{41}\times\sqrt{52}\cos(OAB)$ or $(\sqrt{41})^2=(\sqrt{52})^2+(\sqrt{65})^2-2\times\sqrt{52}\times\sqrt{65}\cos(ABO)$
	M1	For correct substitution into the area of a triangle formula, use their values. e.g. $\frac{1}{2}\times\text{"}\sqrt{41}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}72.349...\text{"}$ or $\frac{1}{2}\times\text{"}\sqrt{65}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}49.184...\text{"}$
	A1	awrt 22
ALT1	M1	For a correct statement for the area, must have $\frac{1}{2}$ (Start with any point, go clockwise or anticlockwise, but must repeat the first point at the end.)
	M1	For a correct evaluation of their determinant, ignore $\frac{1}{2}$
	M1	$\frac{1}{2}$ multiplies their determinant value, correct answer implies this mark.
	A1	Correct **exact** value 22, **not awrt**
ALT2	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For finding half of the perimeter
	M1	Substitute the perimeter and 3 lengths to a correct Heron's formula
	A1	Correct **exact** value 22, **not awrt**
(c)	M1	For $\overrightarrow{OC}=\lambda\overrightarrow{OD}$ (can be implied in their later work)
	M1	For correct vector statement $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}$ oe Can be implied for later substitution of their vectors **Please note**: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{BA}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{AB}$ is also correct, provided they have found a negative value of μ
	ddM1	Equating components and reaching a value for either of their parameters. Dependent on both previous method marks
	A1	Either correct value of their parameters. They do not need to work out both values. Note: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}\Rightarrow\mu=\frac{7}{26}$, $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}\Rightarrow\mu=\frac{19}{26}$
	A1	Correct vector in simplified expression

Type MS · How long is a vector? Magnitudes and finding unknowns

Answer 7 of 7 · Question 8 part a.ii · 3 marks

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Question number	Scheme	Marks